

Sandra L. Perez

Houston, TX | 346-531-5320 | sandrali.perez@outlook.com | [linkedin.com/in/sandralperez](https://www.linkedin.com/in/sandralperez) | saliprued.github.io/portfolio

SUMMARY

Geospatial and Energy Data Professional with 15 years of experience, including more than 10 years supporting the Oil & Gas industry. Experienced in GIS application development, ArcGIS Enterprise, spatial and relational databases, web mapping, dashboards, and geospatial data management. I am currently completing an associate degree in Computer Programming with a specialization in Cloud Computing and Application Development, strengthening my skills in Python, JavaScript, SQL, Git, web development, and cloud-based solutions. I am interested in applying this combined GIS, software development, and energy data experience to utility-scale renewable energy projects.

EDUCATION

Houston City College | Houston, TX |

Graduation: **December 18, 2026**

Degree: *Associate of Applied Science*

- Honors: *Dean's List* **Spring 2024, Fall 2024, Spring 2025, Fall 2025**
- Major: *Computer Programming: Cloud Computing- Application Development Specialization*
- GPA: 3.66

Sergio Arboleda University | Bogotá, Colombia |

Graduated: **08/2019**

Degree: *Master's in Information Management and Geospatial Technologies.*

- Honors: *Summa Cum Laude, Graduate Project awarded "Meritorious".*
- Relevant Coursework: *Geospatial Information Systems, Data Management.*

Francisco José de Caldas District University | Bogotá, Colombia |

Graduated: **11/2009**

Degree: *Graduate Certificate in Geographic Information Systems*

Pedagogical and Technological University of Colombia | Tunja, Colombia |

Graduated: **07/2004**

Degree: *Bachelor's in Geological Engineering*

- Honors: *Undergraduate Research Project awarded "Meritorious".*

TECHNICAL SKILLS

- *Programming & Version Control:* Python, JavaScript, HTML5, CSS3, Node.js, Express, Git, GitHub and Windows.
- *Software Development:* Experience supporting application design, development, testing, deployment, maintenance and documentation in collaborative project environments.
- *Data & Cloud Architecture:* AWS Cloud Computing concepts.
- *Relational & Spatial Databases:* SQL, MySQL, Oracle and PostgreSQL/PostGIS; maintenance, integration and QA/QC.
- *GIS Technologies:* ArcGIS Enterprise, Portal for ArcGIS, ArcGIS Online, ArcGIS Pro, QGIS, ArcGIS Insights.
- *Digital Image Processing:* Proficient in ENVI, SNAP, ERDAS, and Agisoft PhotoScan.
- *Energy & G&G Software:* Technical knowledge of Petrel, DecisionSpace, Openwells, Techlog, and Kingdom Suite.
- *Data Visualization:* Experienced in creating interactive Dashboards and using Power BI for data analysis.
- *Languages:* Native Spanish speaker with Professional Working Proficiency in English (ESL Certified).

TECHNICAL PROJECTS

Flood Risk and Emergency Facilities GIS Analysis: Developed a Python workflow to retrieve, filter, and process geospatial data from ArcGIS REST feature services. Integrated flood hazard, community vulnerability, hospital, and emergency shelter datasets into an ArcGIS Online web map. Identified hospitals and shelters located within the project's flood-risk area to support emergency planning and infrastructure resilience.

College Management System: Developed a Node.js, Express, and MySQL application for managing students, faculty, courses, users, and enrollment records.

AWS Cloud Monitoring Project: Configured EC2 and CloudWatch services, analyzed application logs, created a dashboard, and resolved configuration issues.

PROFESSIONAL EXPERIENCE

Realidad 5.0

02/2022– 08/2023

GIS Developer

- Developed front-end components using Angular for the "RealGIS" application, enabling real-time geographic data collection and visualization.
- Managed corporate architectures for ArcGIS Enterprise web applications, including Portal and ArcGIS Online environments.
- Led the technical integration of complex geographic databases with alphanumeric cadastral systems to ensure data synchronization and integrity.
- Designed interactive dashboards and query viewers to improve decision-making processes for commercial and real estate projects.

Servicio Geologico Colombiano

04/2021 – 12/2021

GIS Professional

- Structured spatial databases for the "Evaluation of the Colombian Paleozoic Basin" project, specifically targeting hydrocarbon exploration data.
- Automated the upload and management of over 200 thematic information layers for urban geoviewers, ensuring data accuracy.
- Produced high-precision structural contour maps and stratigraphic columns to support regional geological assessments.

Agencia Nacional de Hidrocarburos (ANH)

08/2019 – 12/2019

Data Management Consultant

- Collaborated with the data management team to consolidate technical information packages for permanent oil area allocation and bidding rounds.
- Contributed to the development of the official manual for delivering geological and geophysical information to the Colombian Geological Service.
- Ensured QA/QC standards for technical exploration and production (E&P) data provided to international oil companies.

Colombian Data Bank Oil & Gas (Outsourcing)

06/2015 – 02/2018

Data Management Professional

- Administered and updated massive seismic datasets and well logs to support seismic interpreters in evaluating new exploratory opportunities.
- Facilitated the sale and provision of technical exploration information to major oil operating companies, maintaining high service standards.
- Streamlined the internal data retrieval processes, reducing response times for technical information requests from stakeholders.

Halliburton Latin America (Landmark)

06/2012 – 02/2015

Data Management Consultant

- Managed and updated seismic information and well logs to support interpreters in evaluating new exploratory opportunities.
- Supported oil operating companies in the acquisition, sale, and provision of technical E&P information.
- Led the team responsible for managing regional project information in the Caribbean and Llanos areas for ECOPETROL S.A.

AWARDS & ACTIVITIES

- 2026 All-Texas Academic Team
- 2026 World Impact Scholar
- Certified Member, Community Emergency Response Team (CERT)
- Member of the Phi Theta Kappa Honors Society

PUBLICATIONS

Peer-Reviewed Publication

12/2021

Urban growth modeling: Tunja 2017–2035 (own translation).

Bitacora Urbano Territorial

<https://doi.org/10.15446/bitacora.v32n1.87758>